OVAL (ours)
retrieval baseline
eviction († upper bound)
random control
oracle ceiling
dense attention
Pareto-optimal ★
avg@8 accuracy (%)
KV memory retained (KiB per KV head per layer)
OVAL Retv.+Comp.
OVAL Retv.
TOVA †
4-bit keys
Loki r16
H2O †
ExpectedAttn
RaaS
13 methods that store the keys exactly
(all named at right)
OVAL Retv.+Comp. ★ 71.75
OVAL Retv. 71.75
Centroid 71.75
Oracle q(t−1) ★ 71.50
Oracle ★ 71.50
TOVA † 71.50
SparQ 71.50
InfLLM 71.50
RocketKV 71.50
ArkVale 71.25
Quest 71.25
ShadowKV 71.00
Dense 70.75
Loki r16 70.75
4-bit keys ★ 70.75
Random 70.25
FreeKV 70.25
H2O † 68.75
ExpectedAttn 68.00
K-norm 67.75
RaaS 50.50
avg@8 accuracy (%)